Subtraction in $\mathbb{N}$

1. Evaluate: $12 - 5$.
2. Evaluate: $18 - 9$.
3. Calculate: $43 - 18$.
4. Calculate: $61 - 27$.
5. Evaluate: $84 - 39$.
6. Sarah has 95 apples and gives away 42. How many apples does she have left?
7. A bookstore had 135 books and sold 58. How many books remain?
8. A tank contains 240 litres of water, and 115 litres are used. How much water is left?
9. A library has 523 books in a section and checks out 287. How many books are still on the shelf?
10. A company had 1000 pounds and spent 463 pounds. How much budget is remaining?

Solutions

1. **7**
   Explanation: $12 - 5 = 7$.

2. **9**
   Explanation: $18 - 9 = 9$.

3. **25**
   Explanation: Column layout with borrowing:

$$\begin{array}{r} \not{4}^{\,3}\ {}^{1}3 \\ -1\ \ 8 \\ \hline \mathbf{2\ \ 5} \end{array}$$

   Units: Borrow 1 ten ($13 - 8 = 5$). Tens: $3 - 1 = 2$. Total: 25.

4. **34**
   Explanation: Column layout with borrowing:

$$\begin{array}{r} \not{6}^{\,5}\ {}^{1}1 \\ -2\ \ 7 \\ \hline \mathbf{3\ \ 4} \end{array}$$

   Units: Borrow 1 ten ($11 - 7 = 4$). Tens: $5 - 2 = 3$. Total: 34.

5. **45**
   Explanation: Column layout with borrowing:

$$\begin{array}{r} \not{8}^{\,7}\ {}^{1}4 \\ -3\ \ 9 \\ \hline \mathbf{4\ \ 5} \end{array}$$

   Units: Borrow 1 ten ($14 - 9 = 5$). Tens: $7 - 3 = 4$. Total: 45.

6. **53 apples**
   Explanation: Standard column layout without borrowing:

$$\begin{array}{r} 9\,5 \\ -4\,2 \\ \hline \mathbf{5\,3} \end{array}$$

   Units: $5 - 2 = 3$. Tens: $9 - 4 = 5$. Total: 53 apples.

7. **77 books**
   Explanation: Column layout with multiple borrowings:

$$\begin{array}{r} \not{1}^{\,0}\ {}^{1}\not{3}^{\,2}\ {}^{1}5 \\ \quad 5\ \ 8 \\ \hline \mathbf{7\ \ 7} \end{array}$$

Units: Borrow 1 ten ($15 - 8 = 7$). Tens: Borrow 1 hundred ($12 - 5 = 7$). Total: 77 books.

8. **125 litres**
   Explanation: Column layout with borrowing:

$$\begin{array}{r} 2\,\not{4}^{3}\,{}^{1}0 \\ -1\;\;1\;\;5 \\ \hline \mathbf{1\;\;2\;\;5} \end{array}$$

   Units: Borrow 1 ten ($10 - 5 = 5$). Tens: $3 - 1 = 2$. Hundreds: $2 - 1 = 1$. Total: 125 litres.

9. **236 books**
   Explanation: Column layout with multiple borrowings:

$$\begin{array}{r} \not{3}^{4}\,{}^{1}\not{2}^{1}\,{}^{1}3 \\ -2\;\;\;8\;\;7 \\ \hline \mathbf{2\;\;3\;\;6} \end{array}$$

   Units: Borrow 1 ten ($13-7=6$). Tens: Borrow 1 hundred ($11-8=3$). Hundreds: $4-2=2$. Total: 236 books.

10. **537 pounds**
    Explanation: Column layout with borrowing across zeros:

$$\begin{array}{r} \not{1}^{0}\,\not{0}^{9}\,\not{0}^{9}\,{}^{1}0 \\ -\;\;4\;\;6\;\;3 \\ \hline \mathbf{5\;\;3\;\;7} \end{array}$$

    Units: Borrow across zeros ($10 - 3 = 7$). Tens: $9 - 6 = 3$. Hundreds: $9 - 4 = 5$. Total: 537 pounds.